

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**  
**ASSESSMENT TOOL 3 – DEBUGGING & OPTIMISATION TASK**

|                         |                                                                                                                |                      |                                                   |
|-------------------------|----------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------|
| <b>Course Code:</b>     | CSA06                                                                                                          | <b>Course Title:</b> | Design and Analysis of Algorithms                 |
| <b>CO Assessed:</b>     | CO2 – Manipulation (BL1, BL2, BL3)                                                                             | <b>Assessment:</b>   | Assessment Tool 3 – Debugging & Optimisation Task |
| <b>Weightage:</b>       | 10% of CIA                                                                                                     | <b>Total Marks:</b>  | 100                                               |
| <b>CO2 Statement:</b>   | Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3) |                      |                                                   |
| <b>Student Name:</b>    | C.pavani                                                                                                       | <b>Reg. No.:</b>     | 192524260                                         |
| <b>Section / Batch:</b> | CSE [AI&DS]                                                                                                    | <b>Date:</b>         |                                                   |

**Instructions to Students**

- Answer the following question. Total Marks: 100.
- Carefully analyze the given program/code segment before identifying errors.
- Clearly identify logical, syntactical, and performance-related issues in the code.
- Apply systematic debugging techniques to correct the errors effectively.
- Optimize the given solution by improving efficiency, reducing unnecessary computations, or enhancing time complexity wherever possible.
- Provide proper justification for all corrections and optimization decisions.
- Write clean, structured, and well-documented code with appropriate comments.
- Maintain clarity, logical reasoning, and technical accuracy throughout the solution.

| Question Type                 | Focus Area                                                                                      | Questions |
|-------------------------------|-------------------------------------------------------------------------------------------------|-----------|
| Debugging & Optimisation Task | Identify errors, debug programs, and optimize algorithm efficiency using appropriate techniques | Q1        |

**SECTION A – DEBUGGING & OPTIMISATION TASK**

98  
100

**Code of Conduct**

I C.pavani/192524260 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: C.pavani

## RUBRICS FOR CALCULATION

| Criteria               | Wt. | Level 4 – Exemplary                                                                  | Level 3 – Proficient                                   | Level 2 – Developing                          | Level 1 – Beginning                           |
|------------------------|-----|--------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| Problem Identification | 20% | Accurately identifies all errors and root causes with clear understanding            | Identifies most errors with minor gaps                 | Identifies some errors but misses key issues  | Unable to identify errors correctly           |
| Debugging              | 25% | Applies systematic debugging techniques effectively to resolve issues                | Uses appropriate debugging methods with minor errors   | Limited debugging approach; requires guidance | Unable to debug or resolve issues             |
| Optimization           | 20% | Optimizes code by significantly improving time complexity using efficient algorithms | Improves time complexity with minor gaps in efficiency | Limited improvement in time complexity        | No improvement; inefficient approach retained |
| Analysis               | 20% | Demonstrates strong logical reasoning and clear justification of solutions           | Shows reasonable analysis with some justification      | Limited analysis; weak reasoning              | No analytical reasoning demonstrated          |
| Code Quality           | 15% | Produces clean, well-structured code with proper comments and documentation          | Code is mostly clear with minor documentation gaps     | Code lacks structure and proper documentation | Poorly written code with no documentation     |

## Marks Evaluation:

| Criteria               | Wt. | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|------------------------|-----|---------------------|----------------------|----------------------|---------------------|
| Problem Identification | 20% | ✓                   |                      |                      |                     |
| Debugging              | 25% | ✓                   |                      |                      |                     |
| Optimization           | 20% | ✓                   |                      |                      |                     |
| Analysis               | 20% | ✓                   | ✓                    |                      |                     |
| Code Quality           | 15% |                     | ✓                    |                      |                     |
| Total Marks (100):     |     | 98                  |                      |                      |                     |

| Faculty In-charge             | Course Coordinator | Head of Department |
|-------------------------------|--------------------|--------------------|
| Name: <u>Brijender</u>        | Name: _____        | Name: _____        |
| Signature: <u>[Signature]</u> | Signature: _____   | Signature: _____   |
| Date: <u>4/8/26</u>           | Date: _____        | Date: _____        |